



HOSPITAL MANAGEMENT SYSTEM **SOFTWARE TESTING USING SELENIUM WEBDRIVER**

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[1] ABSTRACT AND PROJECT OVERVIEW

The **Hospital Management System (HMS)** is a web-based application designed to manage and automate various activities involved in hospital operations. The system provides a centralised platform for managing patients, doctors, appointments, medical records, billing, and hospital-related information.

In a traditional hospital environment, many activities involve manual record keeping, which can result in data duplication, errors, difficulty in retrieving information, and delays in providing services. The proposed Hospital Management System aims to simplify these processes by providing an organised and user-friendly digital platform.

The system allows authorised users to perform important operations such as patient registration, doctor management, appointment scheduling, patient record management, billing, and viewing medical information. By integrating these functions into a single system, hospital staff can manage information more efficiently.

As this project is developed as a **Software Testing project**, special importance is given to testing the functionality of the web application. **Selenium WebDriver** is used to automate browser-based testing. Selenium allows the team to simulate real user actions such as opening a webpage, entering patient information, clicking buttons, selecting options, logging into the system, and validating the results.

The main objective of the project is to verify that the Hospital Management System works correctly under different conditions and that its major functions provide the expected results. Automated testing helps reduce repetitive manual testing and allows test cases to be executed consistently.

Objectives of the Project

- To develop and test a Hospital Management System.
- To automate important hospital-related operations.
- To verify the correctness of different system modules.
- To identify functional errors and unexpected behaviour.
- To automate web application testing using Selenium.
- To reduce repetitive manual testing.
- To verify that the system behaves correctly for valid and invalid inputs.

Expected Outcome

The expected outcome of the project is a reliable Hospital Management System in which the major functions are tested systematically using Selenium. The testing process should identify defects and ensure that the application performs according to its requirements.

[2] HOSPITAL MANAGEMENT SYSTEM – MODULES

The Hospital Management System can be divided into several important modules, each responsible for a specific set of hospital operations.

1. Login Module

The login module provides authentication for authorised users.

- Username and password entry
- Login validation
- Invalid login handling
- Logout functionality

***Example test** — Enter valid username and password: the user should successfully enter the system.*

2. Patient Management Module

This module manages patient information.

- Add new patient
- View patient details
- Update patient information
- Delete patient records
- Search for patients

Example test — Enter valid patient details: patient record should be successfully created.

3. Doctor Management Module

This module manages information related to doctors.

- Add doctor details
- View doctor information
- Update doctor details
- Search doctors
- Manage doctor availability

4. Appointment Module

The appointment module allows patients or hospital staff to schedule appointments.

- Select patient
- Select doctor
- Select appointment date
- Select appointment time
- Book appointment
- Cancel appointment

Example test — Select a doctor and an available date: appointment should be successfully booked.

5. Medical Records Module

This module stores and manages patient medical information, including:

- Patient history
- Diagnosis
- Treatment details
- Prescription information
- Previous medical records

6. Billing Module

The billing module manages hospital payments and invoices.

- Generate bills
- Add treatment charges
- Add consultation charges
- Calculate total amount
- Display payment information

[3] TESTING USING SELENIUM

Selenium is used to automate the testing of the Hospital Management System. The table below lists the major automated test cases designed for the application, covering both valid and invalid input scenarios.

Test Case	Input	Expected Result
Login with valid details	Correct username/password	Login successful
Login with invalid details	Wrong password	Error message displayed
Add patient	Valid patient details	Patient added
Empty patient form	Empty fields	Validation message
Book appointment	Valid doctor/date	Appointment booked
Search patient	Patient ID	Patient details displayed
Update patient	Modified details	Details updated
Delete patient	Existing patient	Patient removed
Logout	Click Logout	User returned to login page

Testing Objective

The main purpose of using Selenium is to ensure that the important functions of the Hospital Management System work correctly from the user's perspective. Automated test cases can be executed repeatedly whenever changes are made to the application, ensuring consistent regression coverage with minimal manual effort.

Why Selenium Is Used

Selenium is well suited for this project because the Hospital Management System is a web-based application. Instead of manually performing every test, Selenium can automatically perform user actions such as: *Open website* → *Login* → *Enter patient details* → *Submit* → *Verify result*. This entire process can be automated using Selenium, saving effort and improving reliability.

[4] SELENIUM – TOOL INFORMATION

Introduction

Selenium is an open-source automation framework primarily used for testing web applications. It allows testers to automate actions performed in a web browser and verify whether the application behaves as expected. Selenium supports popular browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. For this project, Selenium is used to automate testing of the Hospital Management System.

Important Selenium Components

1. Selenium WebDriver — the main component used to control web browsers. It can:

- Open a browser and open a webpage

- Find webpage elements
- Enter text and click buttons
- Select options and read webpage information
- Close the browser

2. Selenium IDE — a browser-based tool that can record and replay user actions, useful for creating simple automated test cases without writing a large amount of code.

3. Selenium Grid — allows automated tests to be executed across different browsers and environments (e.g. the same HMS test can be executed on Chrome, Firefox, and Edge).

Selenium Workflow in Our Project

1. Launch Browser → 2. Open Hospital Management System → 3. Locate Web Elements → 4. Perform User Actions → 5. Submit Data → 6. Verify Expected Result → 7. Record Test Result

Example Selenium Test

A simple Selenium test can automate the login process:

```
WebDriver driver = new ChromeDriver();

driver.get("http://localhost:8080/hospital");

driver.findElement(By.id("username")).sendKeys("admin");

driver.findElement(By.id("password")).sendKeys("admin123");

driver.findElement(By.id("login")).click();

System.out.println("Login test completed");

driver.quit();
```

Explanation — *ChromeDriver()* opens Google Chrome; *get()* opens the Hospital Management System; *findElement()* finds an element on the webpage; *sendKeys()* enters information into a field; *click()* clicks a button; *quit()* closes the browser.

[5] Conclusion

The Hospital Management System provides a centralised platform for managing important hospital activities such as patient management, doctor management, appointments, medical records, and billing. Testing is essential to ensure that these functions work correctly. Selenium WebDriver provides an effective way to automate web application testing. By creating automated test cases, repetitive testing activities can be performed faster and more consistently. The combination of the Hospital Management System and Selenium testing helps demonstrate the practical application of software testing and test automation in a real-world healthcare system.

[6] Future Scope

- AI-based test case generation to automatically create new Selenium scripts as the application grows.
- Cross-browser testing using Selenium Grid across Chrome, Firefox, and Edge for wider coverage.
- CI pipeline integration so Selenium test suites run automatically on every code change.
- Extending automated coverage to performance and load testing of the system.
- Mobile application testing support using Appium alongside the existing Selenium suite.
- Security testing modules to validate authentication and data-protection mechanisms.